

BFS Traversal

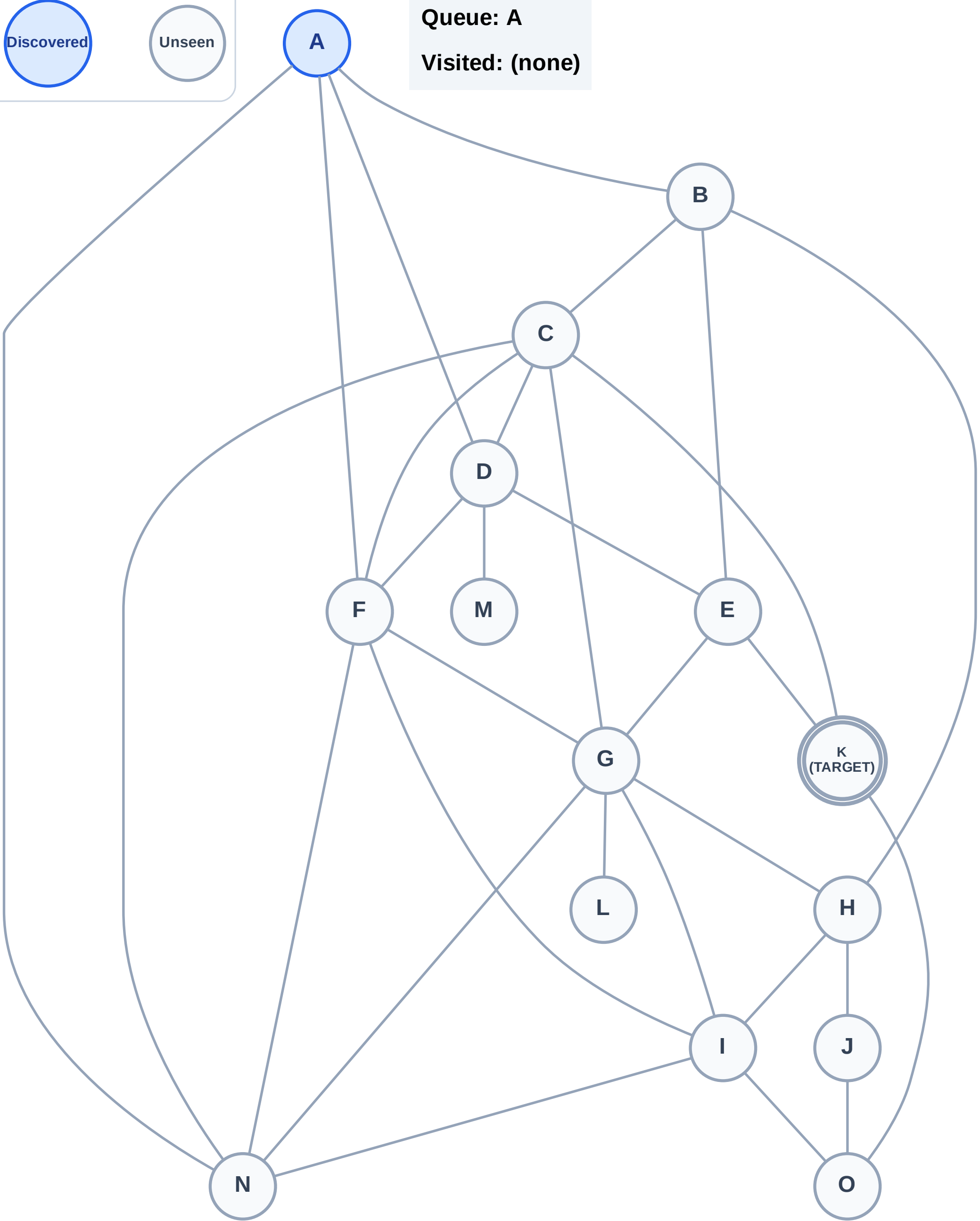
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

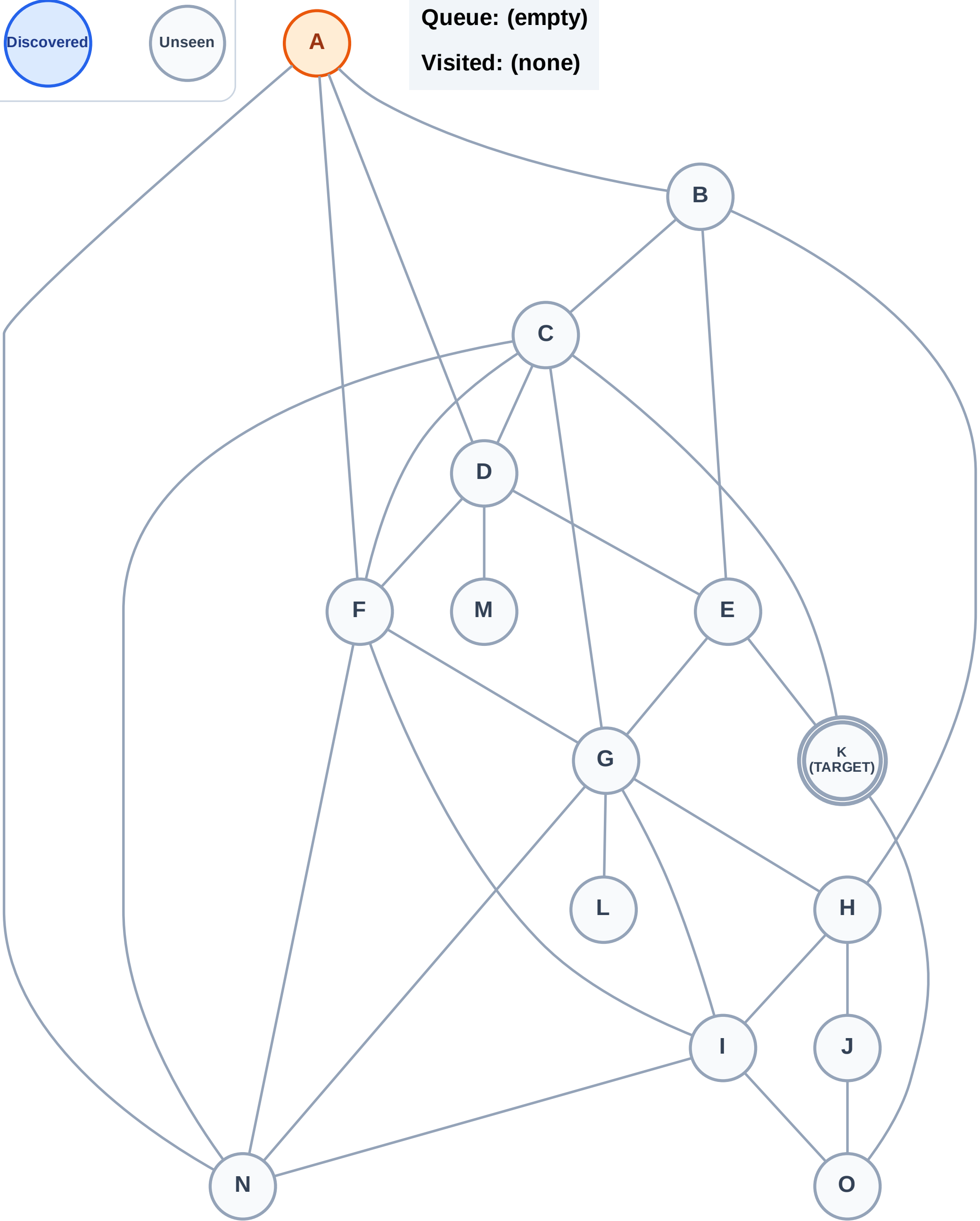
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



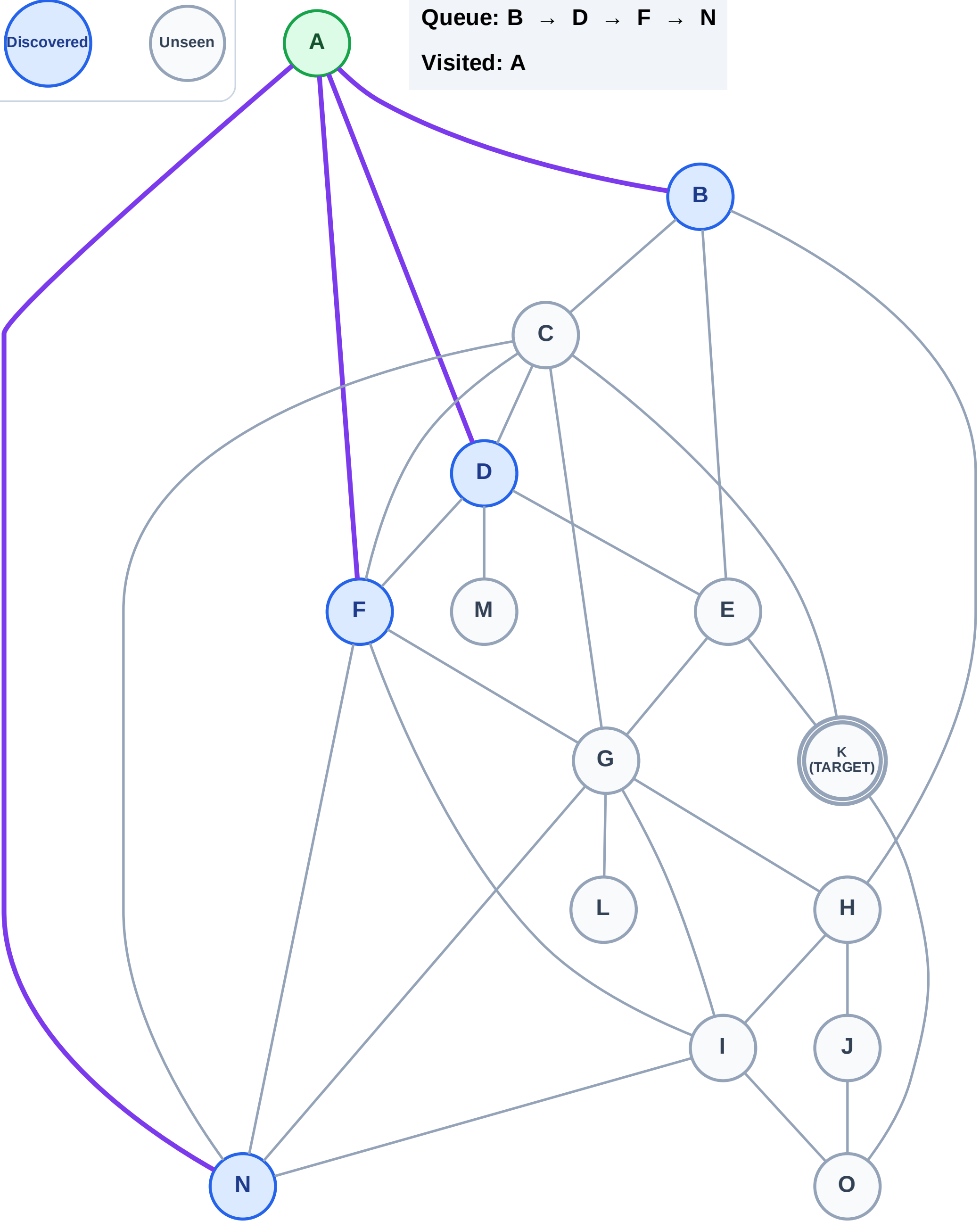
BFS Traversal

Step 3: Discover B, D, F, N from A

Legend



Queue: B → D → F → N
Visited: A



BFS Traversal

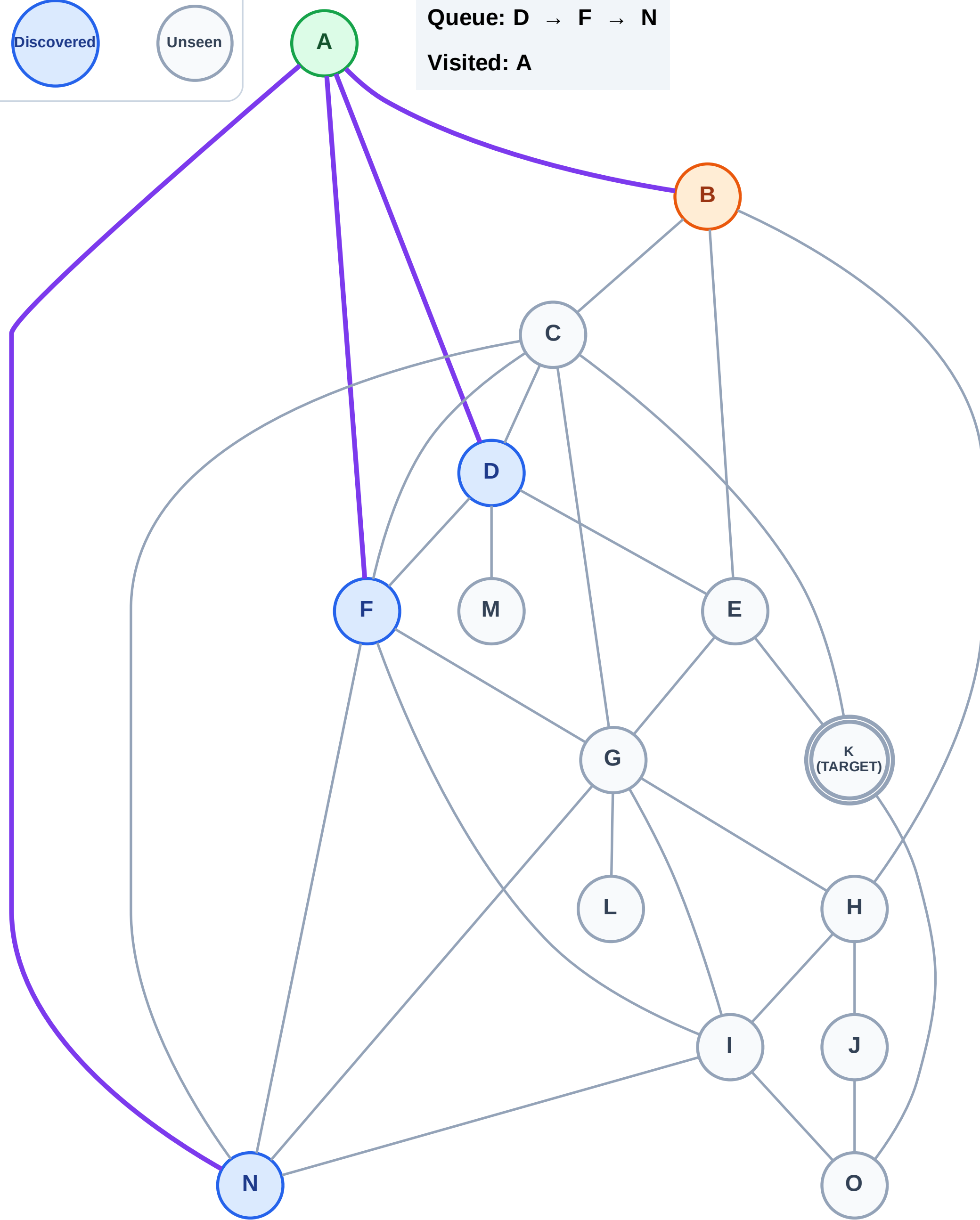
Step 4: Process node B

Legend



Queue: D → F → N

Visited: A



BFS Traversal

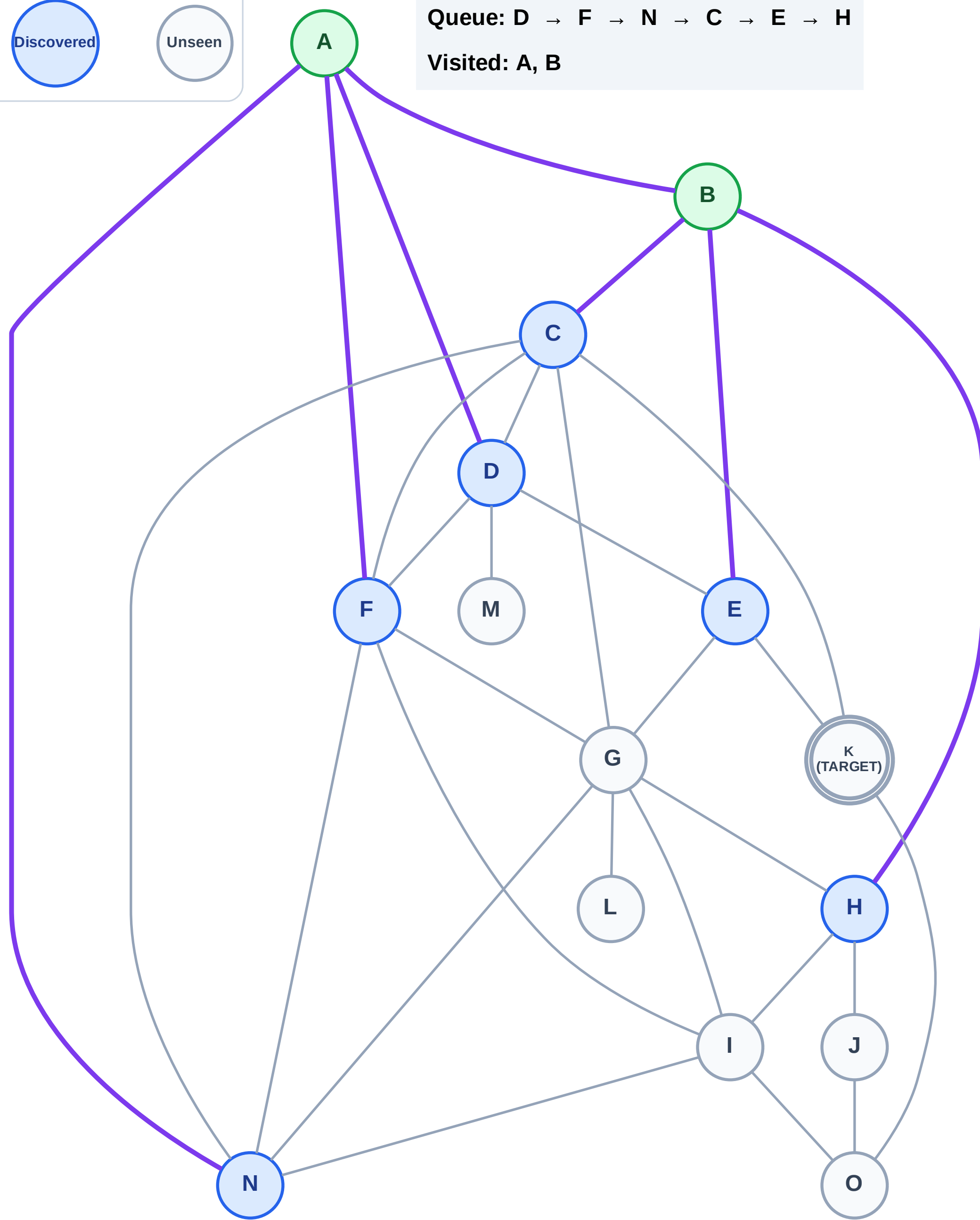
Step 5: Discover C, E, H from B

Legend



Queue: D → F → N → C → E → H

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

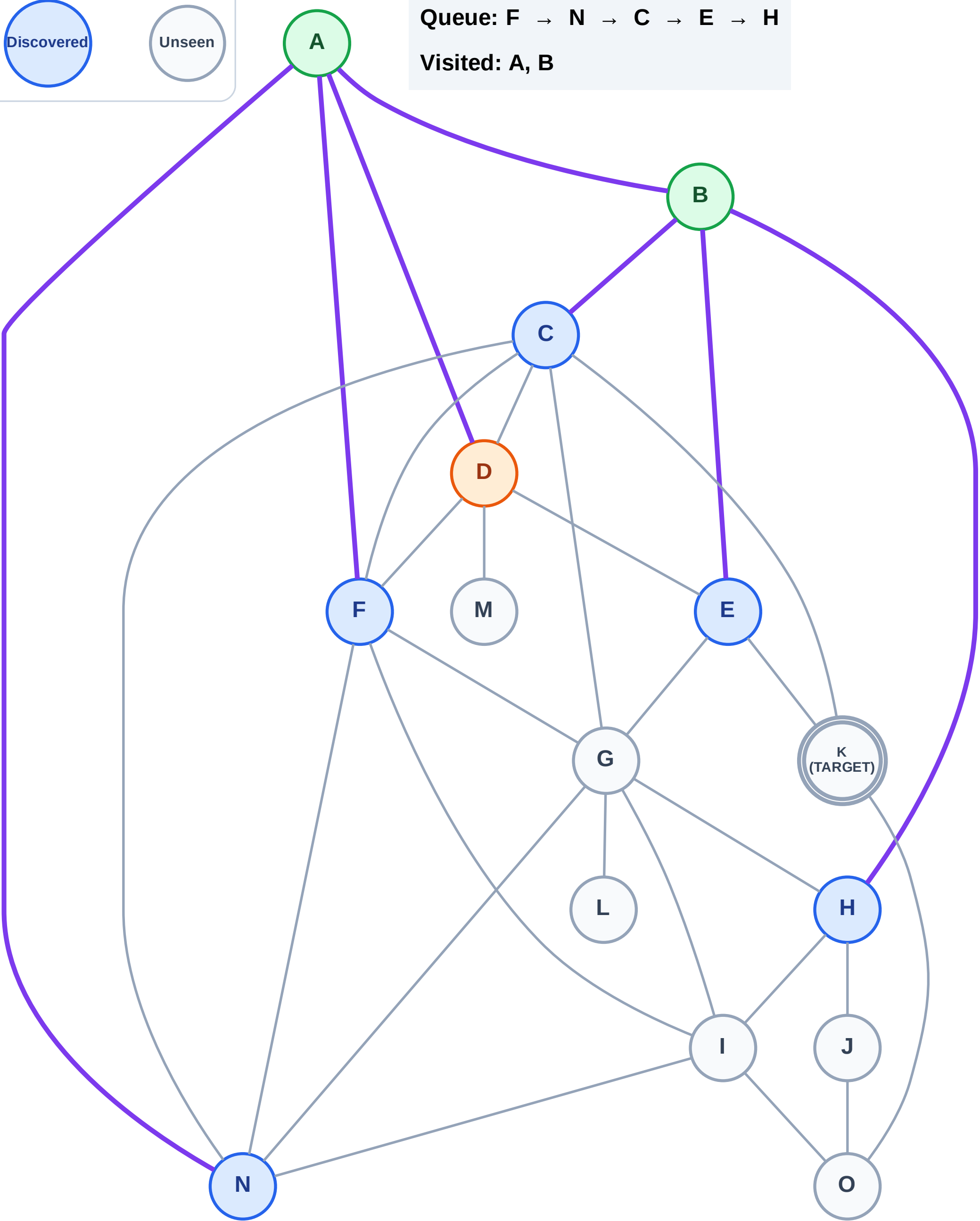
Processing

Discovered

Unseen

Queue: F → N → C → E → H

Visited: A, B



BFS Traversal

Step 7: Discover M from D

Legend

Complete

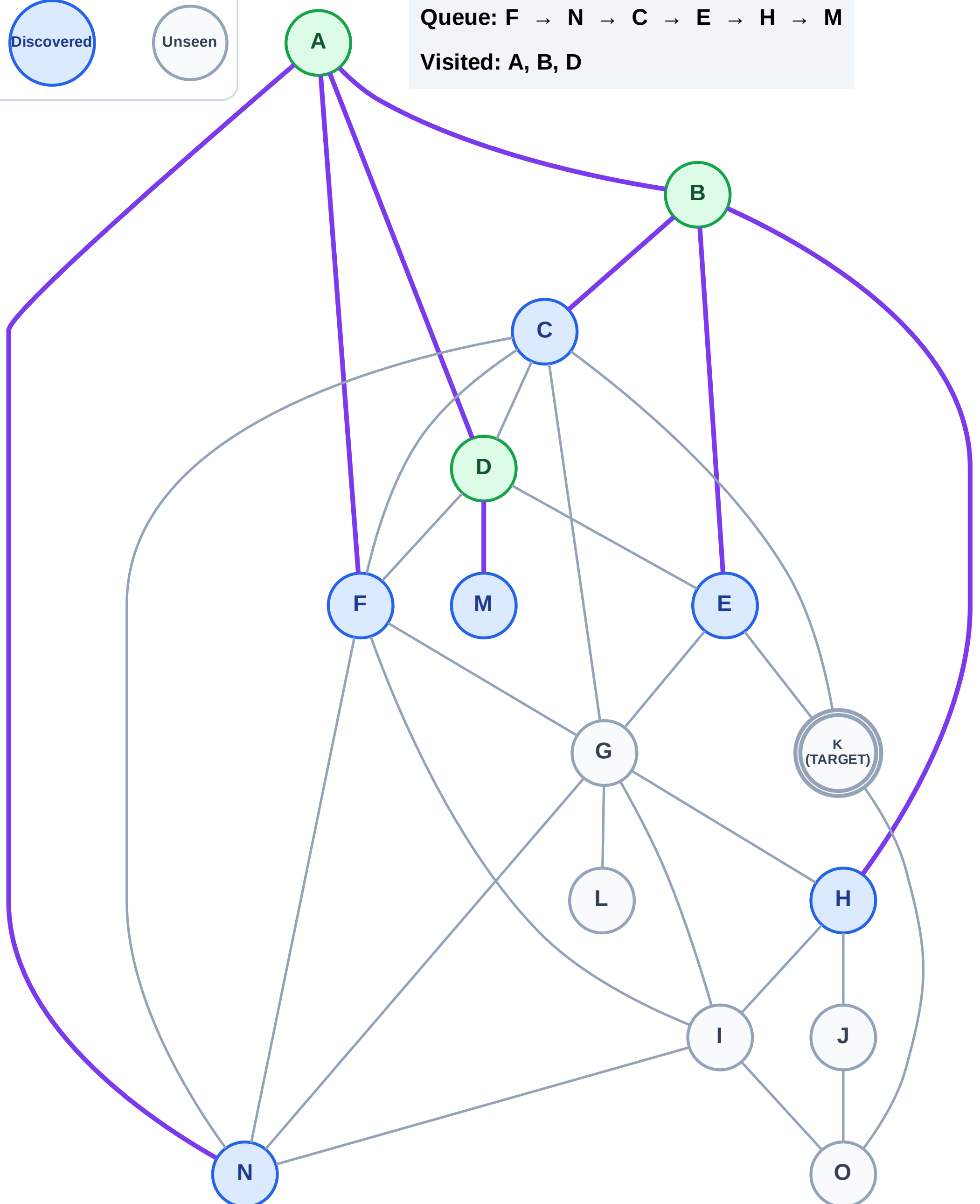
Processing

Discovered

Unseen

Queue: F → N → C → E → H → M

Visited: A, B, D



BFS Traversal

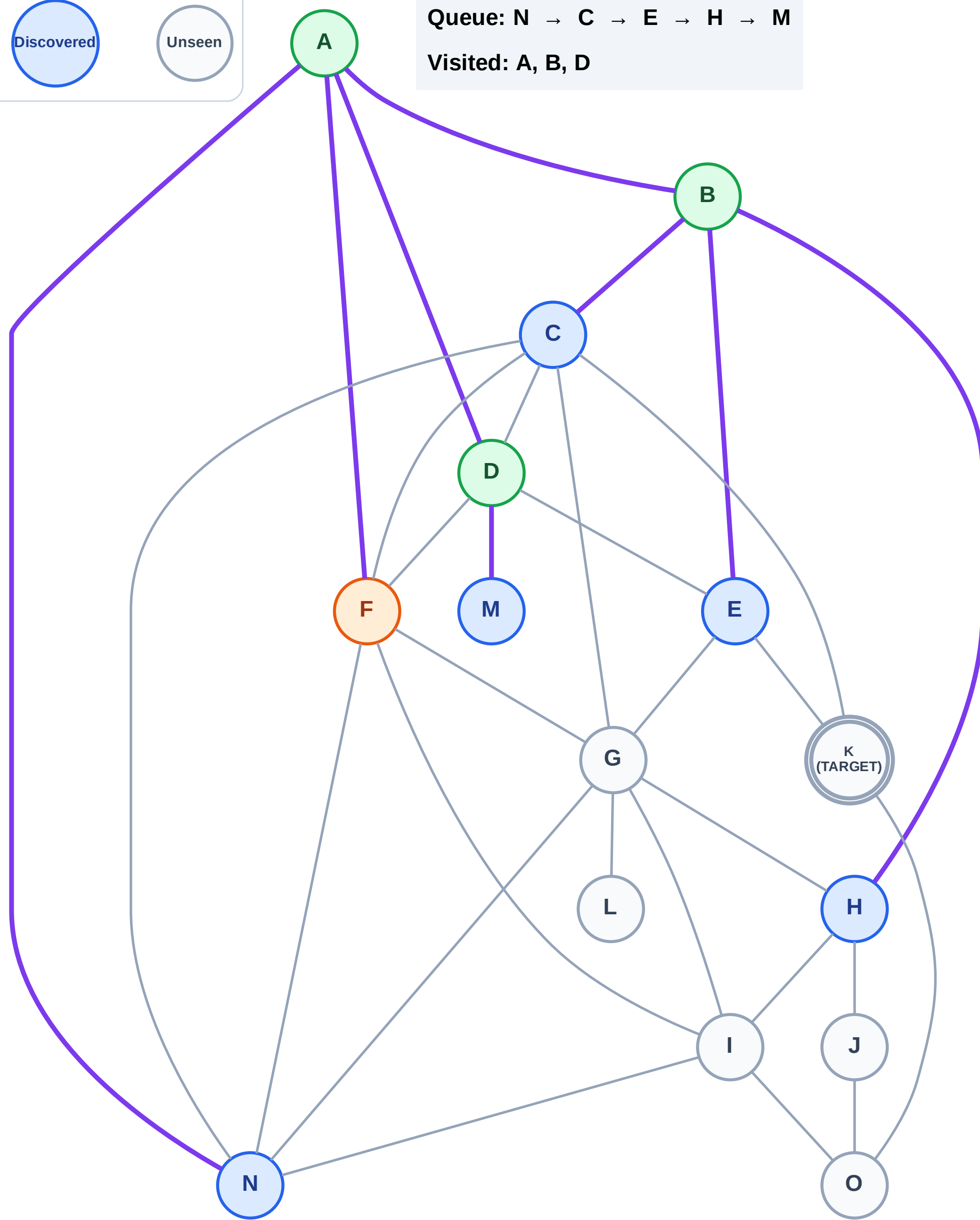
Step 8: Process node F

Legend



Queue: N → C → E → H → M

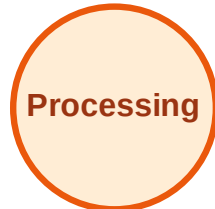
Visited: A, B, D



BFS Traversal

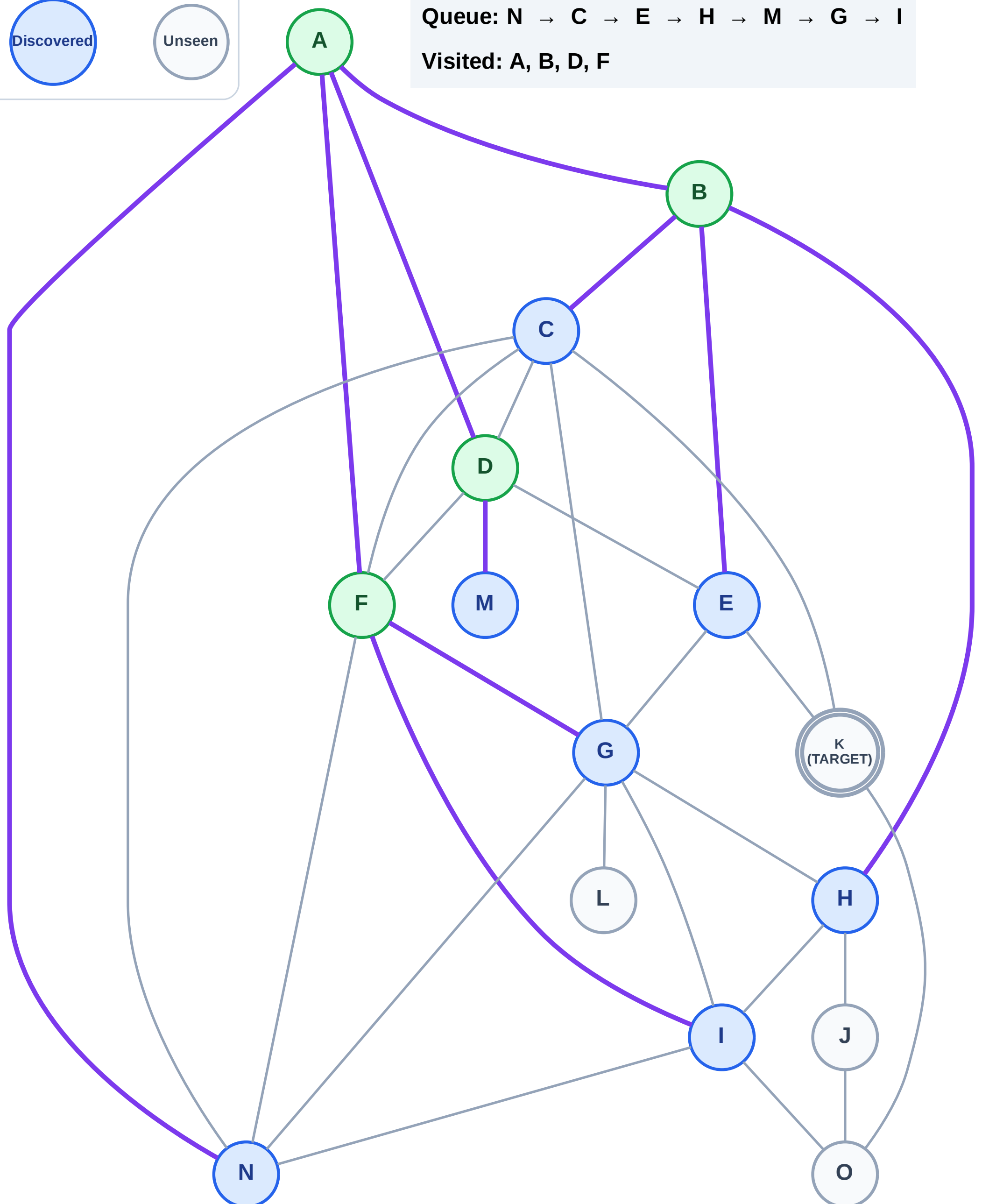
Step 9: Discover G, I from F

Legend



Queue: N → C → E → H → M → G → I

Visited: A, B, D, F



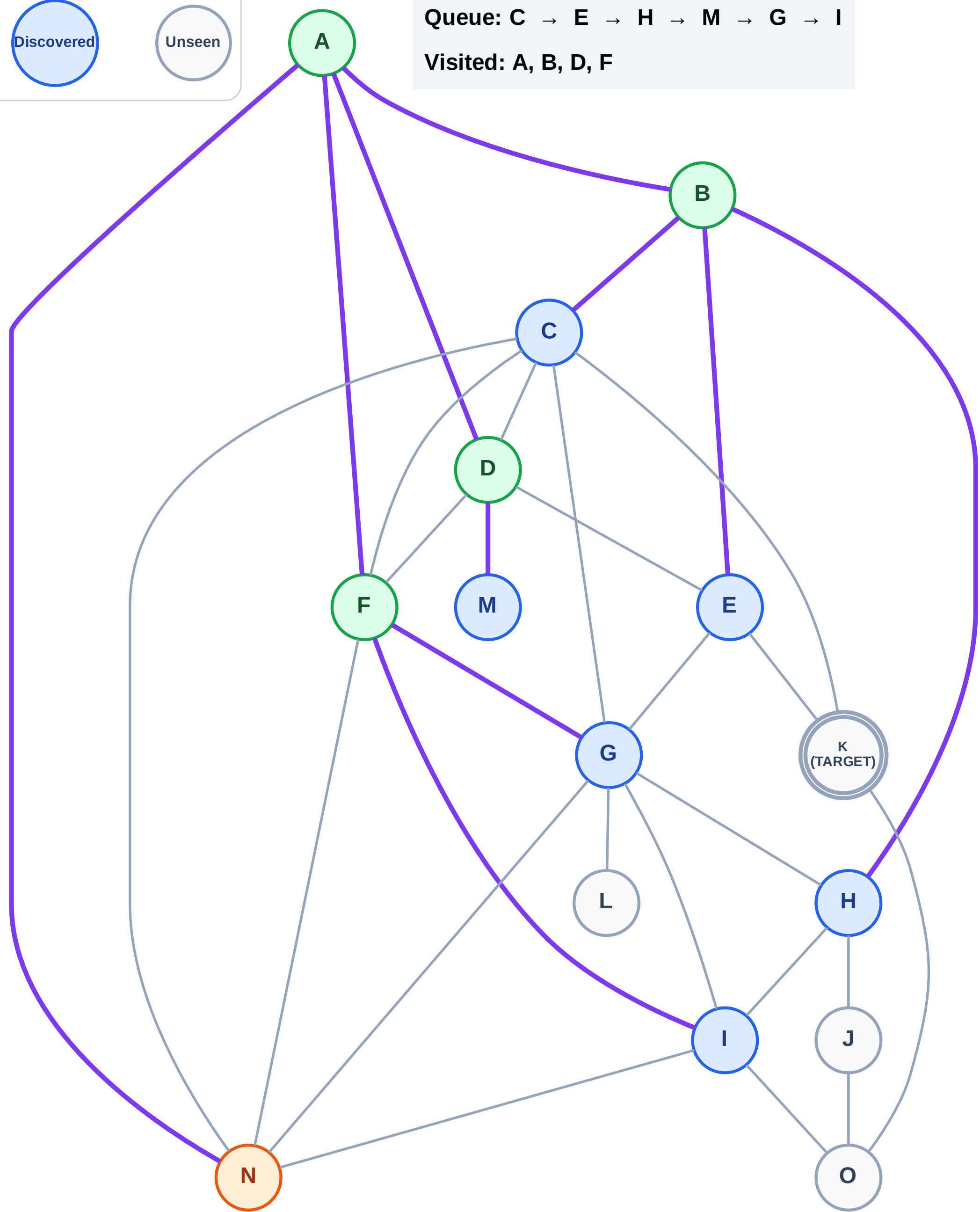
BFS Traversal

Step 10: Process node N

Legend



Queue: C → E → H → M → G → I
Visited: A, B, D, F



BFS Traversal

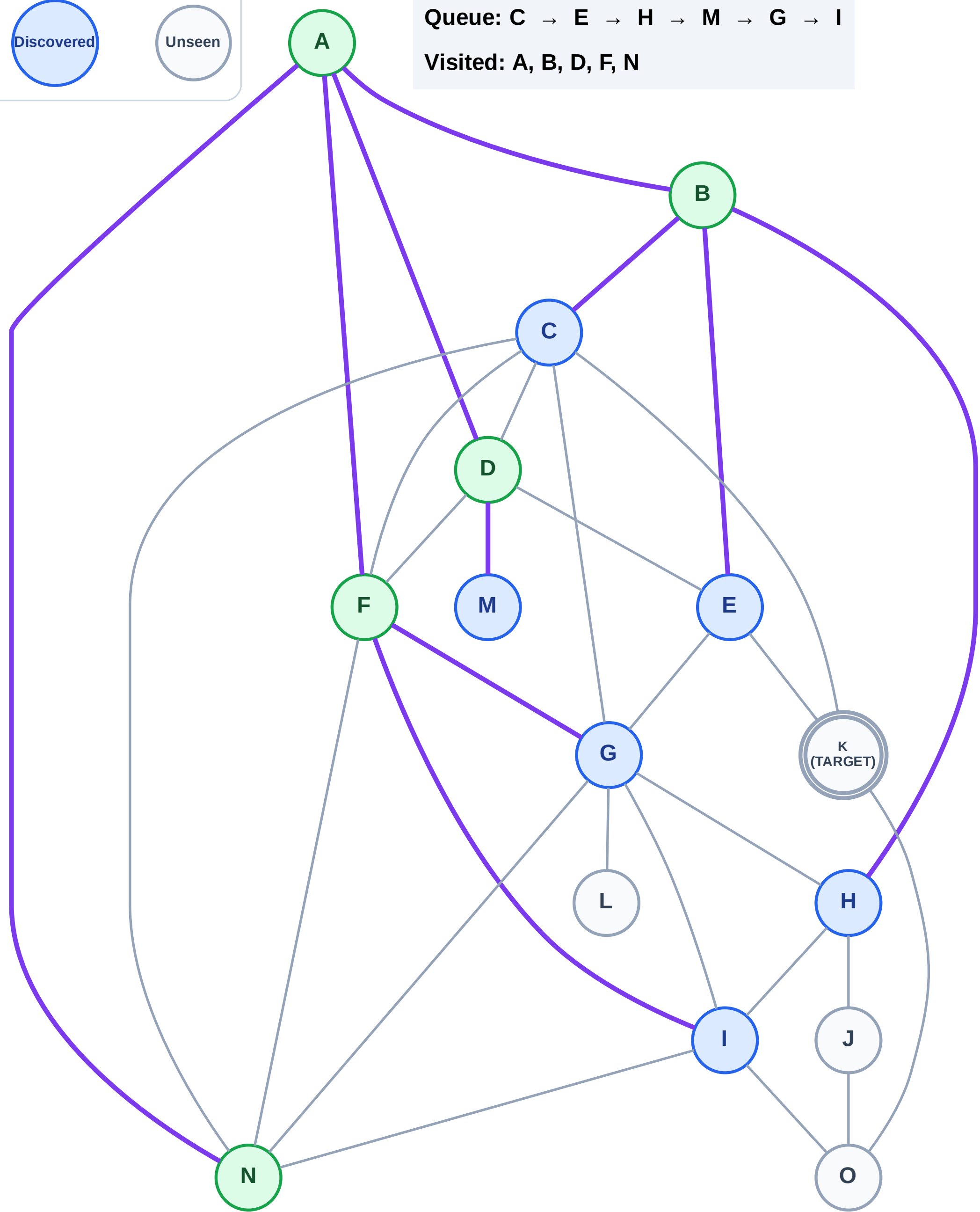
Step 11: No new neighbors from N

Legend



Queue: C → E → H → M → G → I

Visited: A, B, D, F, N



BFS Traversal

Step 12: Process node C

Legend

Complete

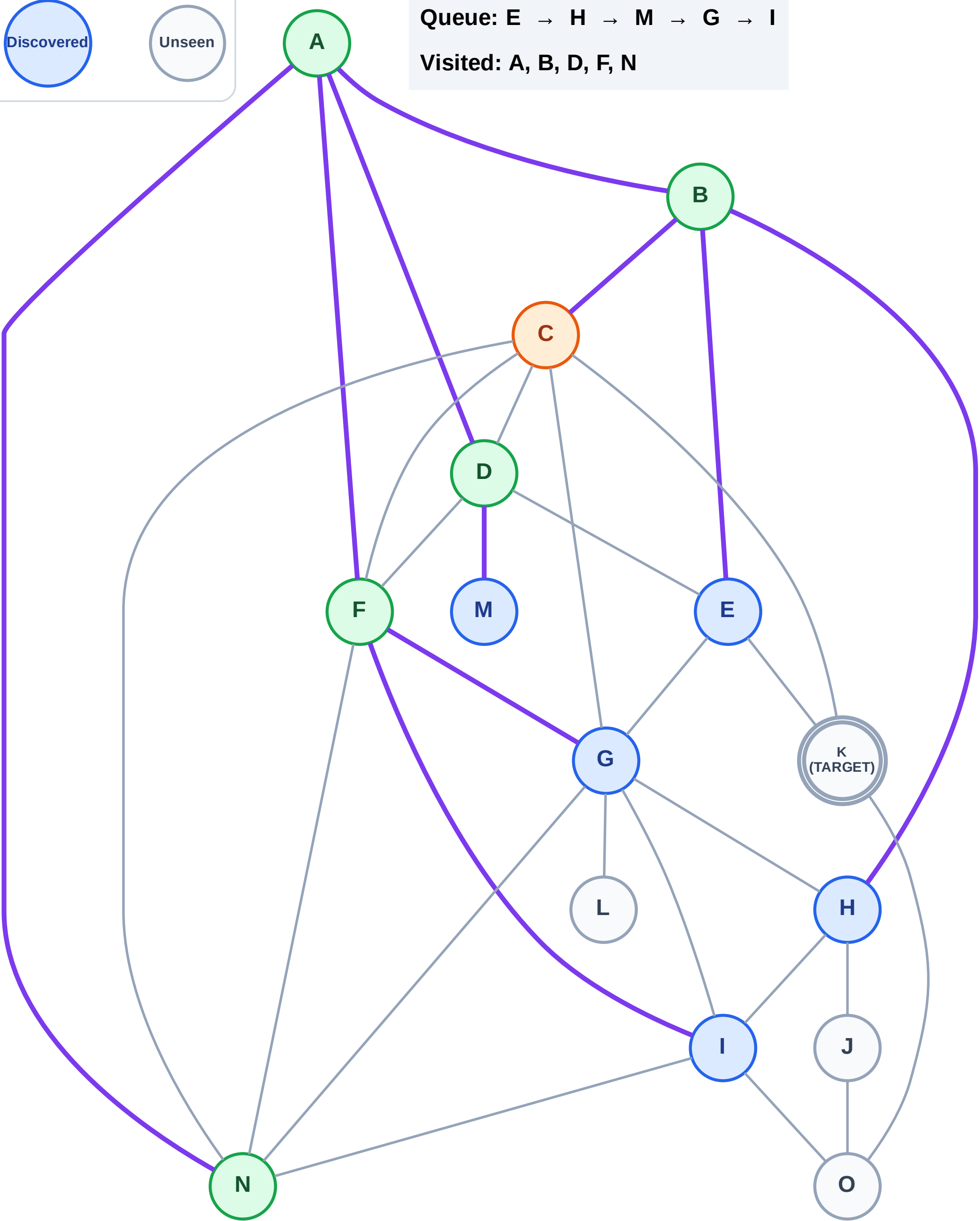
Processing

Discovered

Unseen

Queue: E → H → M → G → I

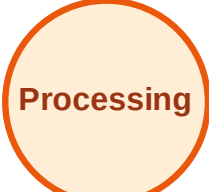
Visited: A, B, D, F, N



BFS Traversal

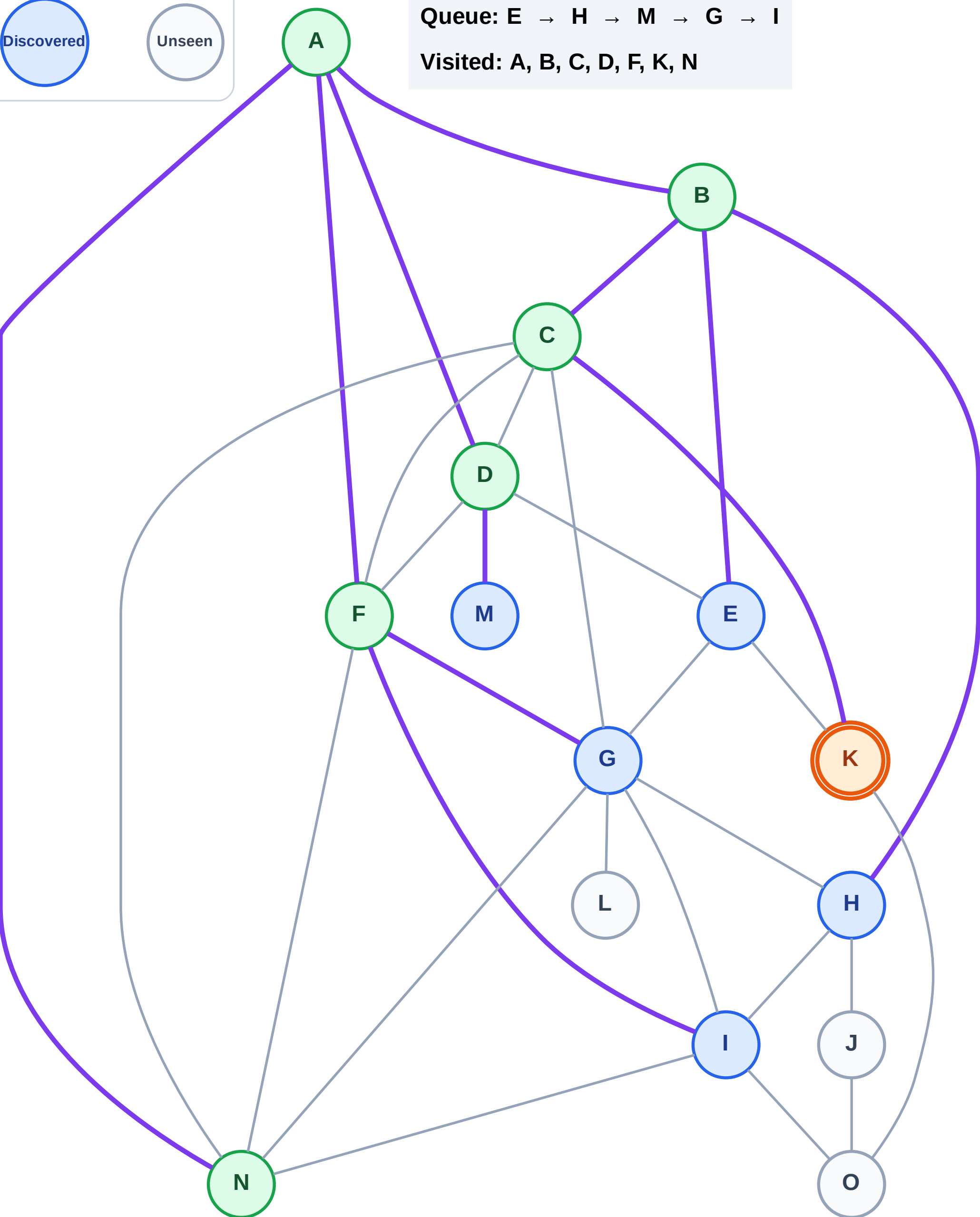
Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend



Queue: E → H → M → G → I

Visited: A, B, C, D, F, K, N



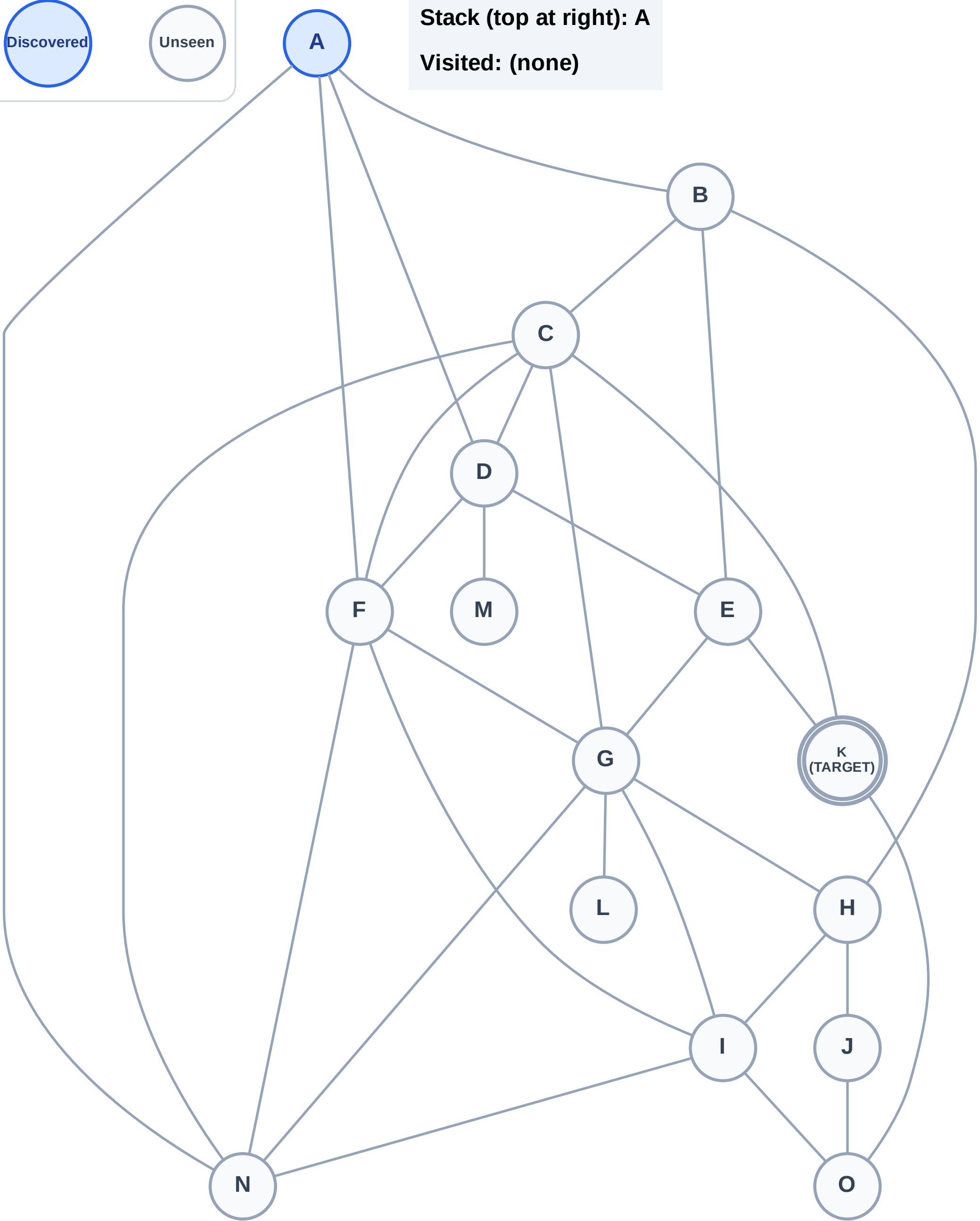
DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

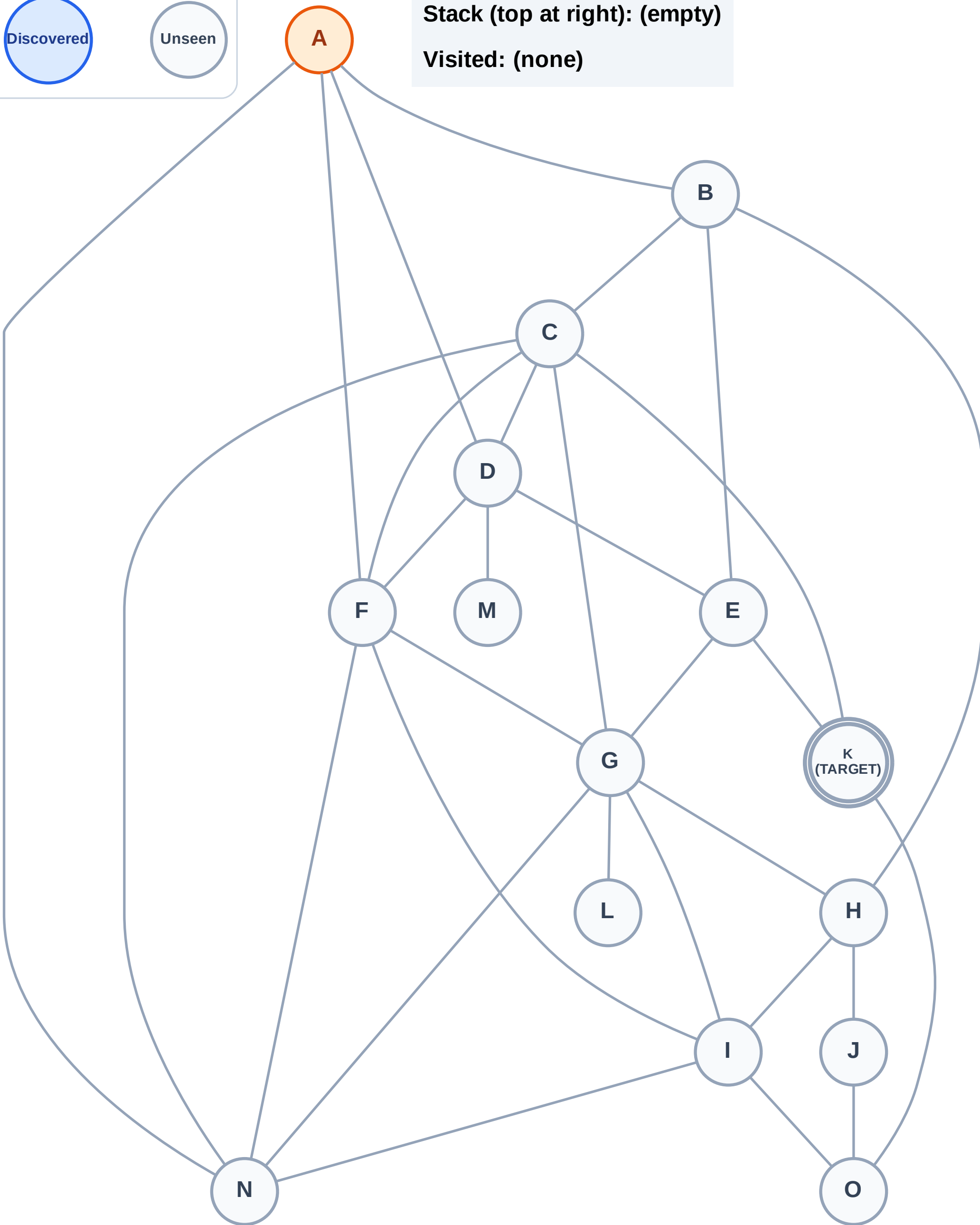
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)

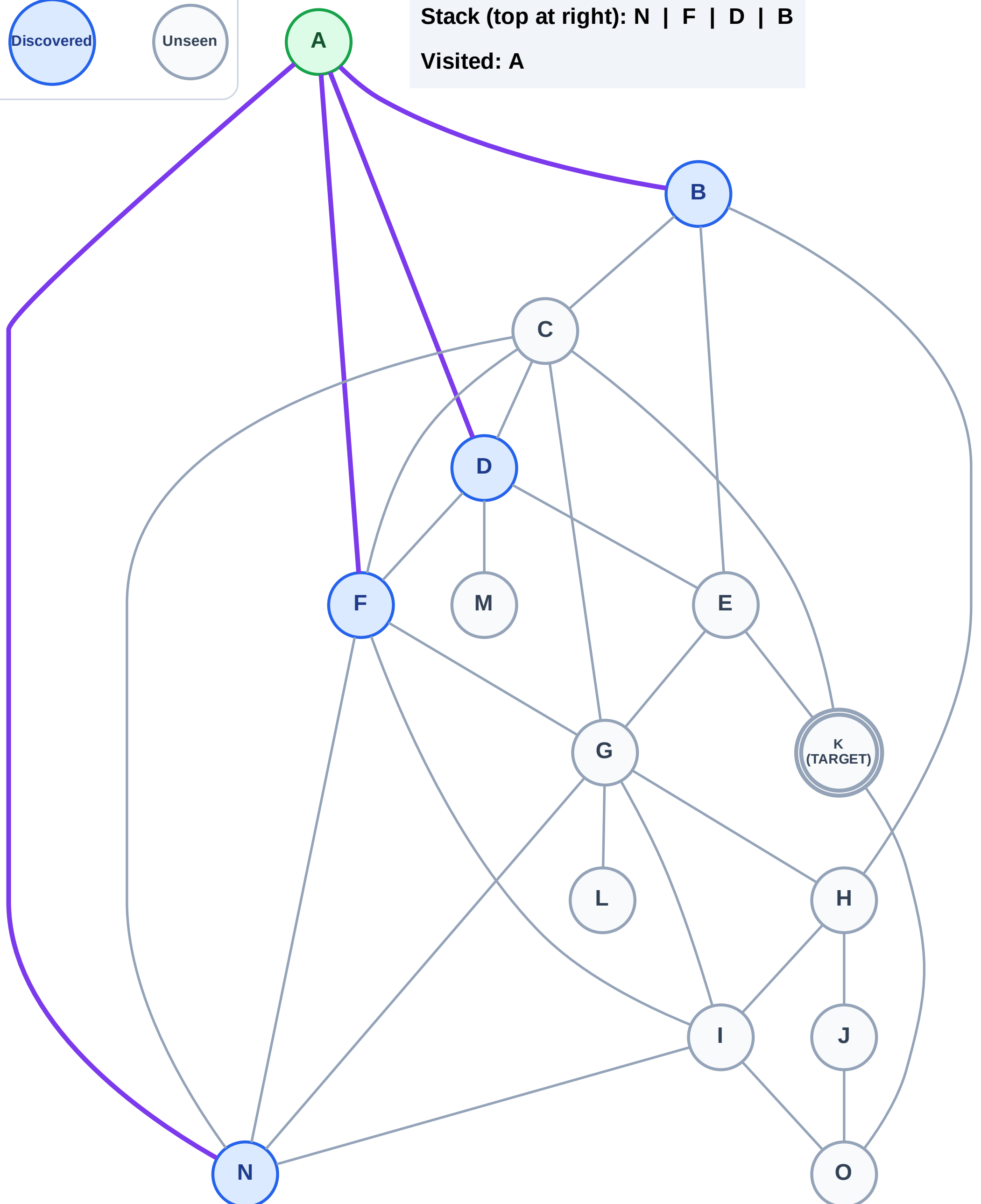


Step 3: Discover N, F, D, B from A

Stack (top at right): N | F | D | B

Legend

Unseen



DFS Traversal

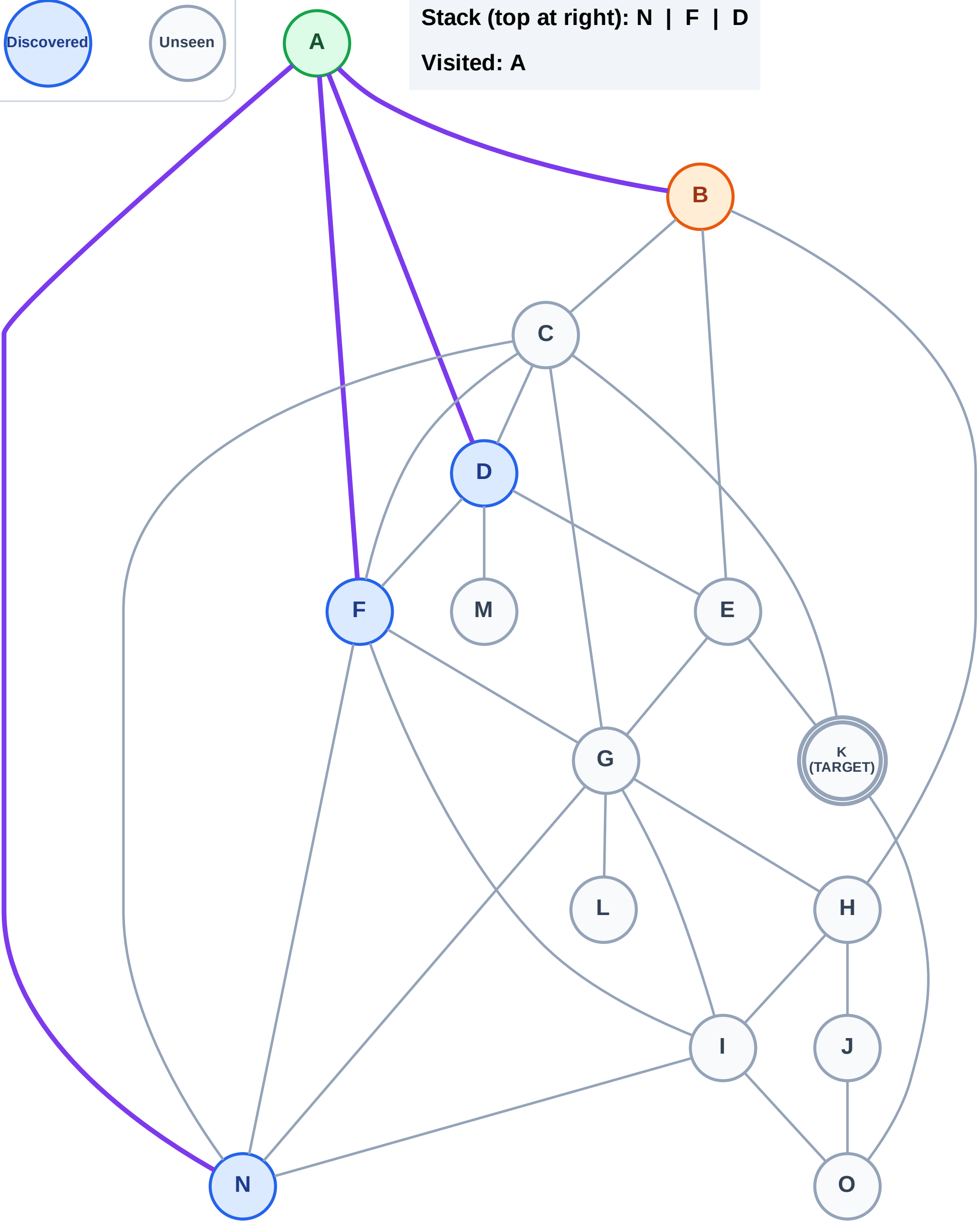
Step 4: Process node B

Legend



Stack (top at right): N | F | D

Visited: A



DFS Traversal

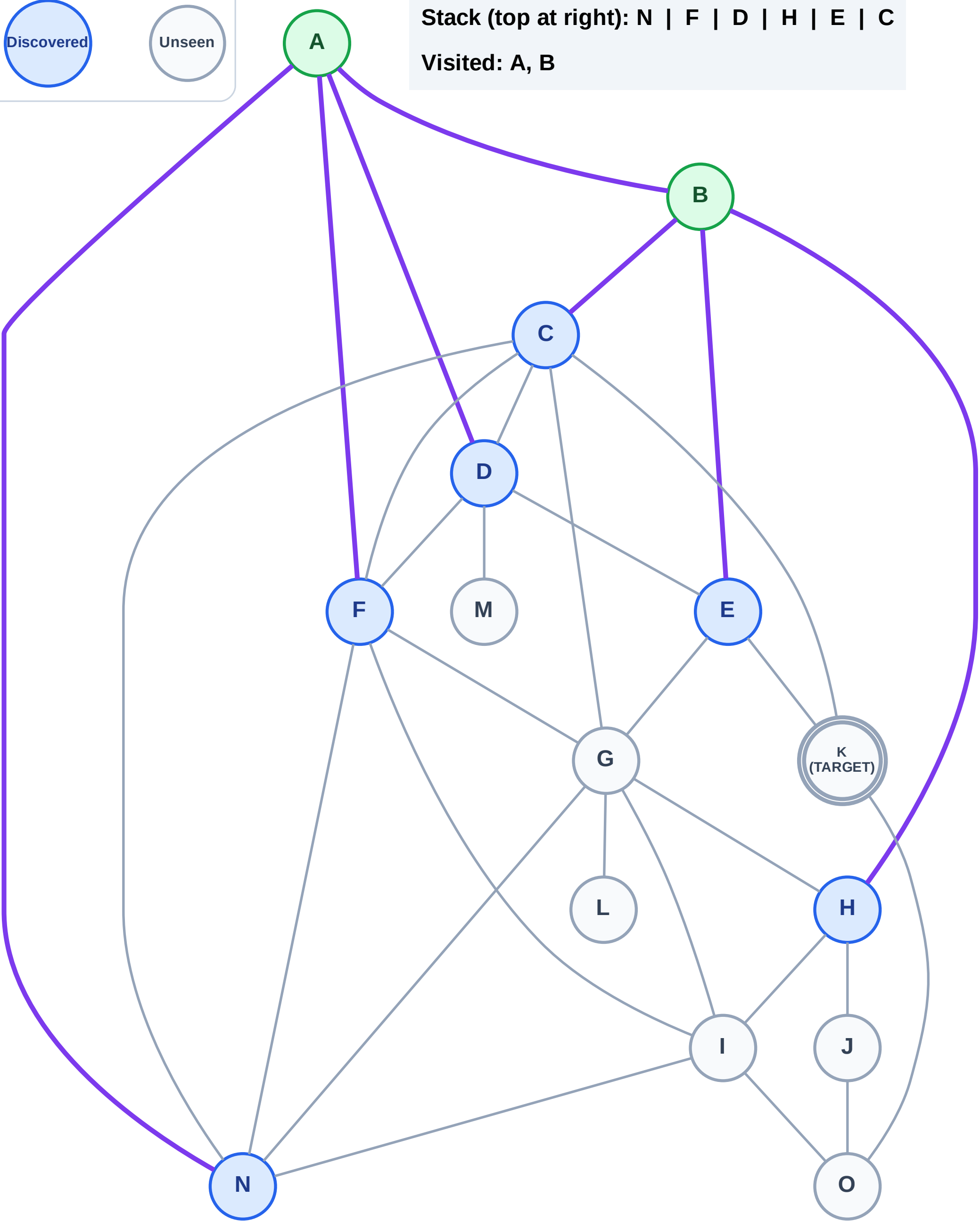
Step 5: Discover H, E, C from B

Legend



Stack (top at right): N | F | D | H | E | C

Visited: A, B



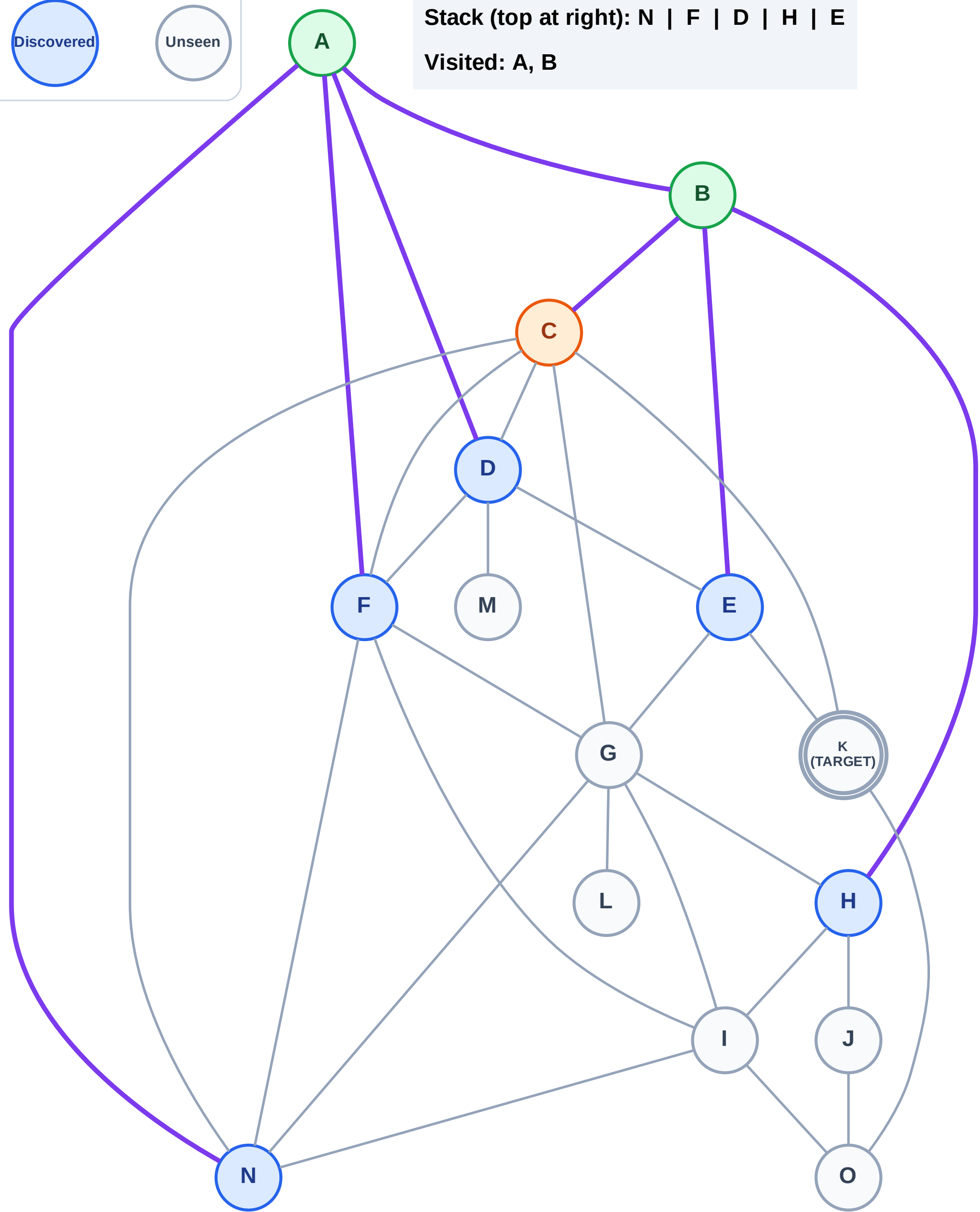
DFS Traversal

Step 6: Process node C

Legend



Stack (top at right): N | F | D | H | E
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | F | D | H | E

Visited: A, B, C, K

